VPN connected but internal sites don't load: DNS and split tunnelling

Article KB0010018 | Owner: Network Support | Subcategory: VPN | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS laptops (MacBook Pro/Air), GlobalProtect VPN client version 6.x or later

Summary

When the GlobalProtect VPN reports a successful connection but internal Acme sites remain unreachable, the problem is usually caused by DNS resolution failures or incorrect split-tunnelling settings. This article walks users through flushing the DNS cache, confirming split-tunnel configuration, and checking the VPN certificate and client version. Service-desk agents can follow a separate procedure to validate certificates, update the client, and adjust split-tunnel policies.

Symptoms

- VPN client shows "Connected" status but internal sites such as intranet.acme.example, sharepoint.acme.example, or internal file shares fail to load.
- Web browsers display DNS errors like "DNS_PROBE_FINISHED_NXDOMAIN" or "Server DNS address could not be found".
- Pinging internal IP addresses (e.g., 10.0.0.5) times out or returns "Request timed out".
- nslookup for an internal host returns a non-authoritative answer from an external ISP DNS server.
- Split-tunnel status in the GlobalProtect tray icon shows "Off" or displays unexpected network routes.

Common causes

- Expired or corrupted GlobalProtect certificate on the laptop, causing the tunnel to fall back to a less-secure mode that blocks corporate DNS.
- Split-tunnelling misconfigured so that DNS queries are sent to the local ISP's resolvers instead of Acme's internal DNS servers.
- Stale DNS cache on the client machine retaining outdated or incorrect records.
- Outdated GlobalProtect client version lacking recent fixes for DNS handling or certificate renewal.
- Local home router or personal firewall blocking UDP/TCP port 443 or interfering with DNS traffic after the VPN is up.

Preliminary Checks

Before proceeding with any changes, verify that the VPN client indeed reports a connected state and note the displayed virtual adapter IP address. Open a command prompt (Windows) or terminal (macOS) and try to ping an internal resource by its IP address (for example, 10.10.20.5). If the ping succeeds, the issue is likely name-resolution related; if it fails, the tunnel may not be routing traffic correctly.

Next, check which DNS servers the VPN adapter is using. On Windows run ipconfig /all and look for the "DNS Servers" line under the GlobalProtect adapter; on macOS use scutil --dns or view Network preferences. If the listed servers are public ISP addresses (e.g., 8.8.8.8) instead of Acme's internal DNS (typically 10.0.0.10/10.0.0.11), split-tunnelling is misconfigured.

Self-Help Procedure: Flush DNS and Verify Split-Tunnel Settings

Follow these steps to clear the local DNS cache and confirm that split-tunnelling is set to send corporate DNS queries through the VPN tunnel. Performing these actions often restores access to internal sites without needing administrator intervention.

1. Disconnect the GlobalProtect VPN if currently connected, then reconnect to ensure a fresh tunnel.
2. On Windows: open Command Prompt as Administrator and run ipconfig /flushdns. On macOS: open Terminal and run sudo killall -HUP mDNSResponder (enter your password when prompted).
3. After flushing, display the VPN adapter's DNS servers again (ipconfig /all on Windows, scutil --dns on macOS). Verify that the servers listed are Acme's internal DNS addresses.
4. If the DNS servers are still external, open the GlobalProtect client settings (gear icon -> Settings -> Advanced) and ensure that "Enable split tunneling" is checked and that the corporate DNS suffixes (e.g., acme.example) are included in

the split-tunnel list.
5. Re-test connectivity by opening a browser to an internal site (intranet.acme.example) or by pinging an internal hostname. Success indicates the issue is resolved.

Service Desk Procedure: Validate VPN Certificate and Client Version

Agents should confirm that the laptop's GlobalProtect certificate is valid and that the client is running a supported version. An expired or corrupted certificate can cause the tunnel to route traffic incorrectly, leading to DNS failures.

1. Ask the user to provide the GlobalProtect client log (right-click the tray icon -> Diagnostics -> Get Logs) and locate the certificate expiration date in the log or in Certificates Manager (certmgr.msc).
2. If the certificate has expired or is marked as untrusted, coordinate with the Network Support team to issue a replacement certificate via the Internal CA and push it to the device through Intune.
3. Check the installed GlobalProtect version (client -> About). Compare it against the approved baseline (currently 6.2.4). If older, instruct the user to upgrade via Company Portal or provide the latest installer from the internal software repository.
4. After certificate renewal or client upgrade, have the user disconnect and reconnect the VPN, then repeat the DNS flush and verification steps from the self-help procedure.
5. Document the certificate expiry date, client version, and any configuration changes made in the ticket.

Verification and Confirmation

Once the above actions are completed, confirm that internal resources are reachable and that the VPN is operating with correct DNS settings. This step ensures the problem is truly resolved before closing the ticket.

1. Have the user attempt to access multiple internal services (intranet, SharePoint, internal file share) and note any lingering errors.
2. Run nslookup for an internal hostname and confirm the response comes from an internal DNS server (10.0.0.10/10.0.0.11).
3. Ping an internal IP address to verify layer-3 connectivity.
4. If all tests succeed, close the ticket with a resolution note summarizing the actions taken (e.g., "Flushed DNS, corrected split-tunnel, renewed VPN certificate, updated client to 6.2.4").

When to escalate

If the user still cannot reach internal sites after completing the self-help and service-desk procedures, escalate to the Network Support team. Include the following details in the escalation: GlobalProtect client logs showing any connection or DNS errors, screenshot of the VPN adapter's DNS servers, certificate expiry date and thumbnote, current GlobalProtect version, and a description of the home network environment (router model, any VPN passthrough or firewall settings). Network Support will examine split-tunnel policies on the GlobalProtect gateway, verify corporate DNS forwarding rules, and, if necessary, push a corrected configuration profile via Intune.

Keywords: VPN connected but no intranet, GlobalProtect DNS issue, split tunnelling not working, internal sites not loading over VPN, flush DNS VPN, VPN certificate expired, GlobalProtect client update, DNS_PROBE_FINISHED_NXDOMAIN, cannot reach internal shares, VPN split tunnel configuration